

I) ECUACIONES LINEALES DE PRIMER ORDEN

Ejercicio 1

$$y' + y \cos x = \cos x$$

$$\mu = e^{\int \cos x \, dx}$$

$$\mu = e^{\operatorname{sen} x}$$

$$e^{\operatorname{sen} x} y' + e^{\operatorname{sen} x} y \cos x = e^{\operatorname{sen} x} \cos x$$

$$\frac{d}{dx} (ye^{\operatorname{sen} x}) = e^{\operatorname{sen} x} \cos x$$

$$ye^{\operatorname{sen} x} = \int e^{\operatorname{sen} x} \cos x \, dx + C$$

$$ye^{\operatorname{sen} x} = e^{\operatorname{sen} x} + C$$

$$y = \frac{e^{\operatorname{sen} x} + C}{e^{\operatorname{sen} x}}$$

$$y = 1 + Ce^{-\operatorname{sen} x}$$

$$\boxed{y = 1 + Ce^{-\operatorname{sen} x}}$$

Operación auxiliar

$$I = \int e^{\operatorname{sen} x} \cos x \, dx$$

$$u = \operatorname{sen} x$$

$$du = \cos x \, dx$$

$$I = \int e^u \, du$$

$$I = e^u + C$$

$$I = e^{\operatorname{sen} x} + C$$

Ejercicio 2

$$y' + 2y = x^2 + 2x$$

$$\mu = e^{\int 2 dx}$$

$$\mu = e^{2x}$$

$$e^{2x}y' + 2e^{2x}y = e^{2x}(x^2 + 2x)$$

$$\frac{d}{dx}(ye^{2x}) = e^{2x}(x^2 + 2x)$$

$$ye^{2x} = \int e^{2x}(x^2 + 2x) dx + C$$

$$ye^{2x} = e^{2x} \left(\frac{x^2}{2} + \frac{x}{2} - \frac{1}{4} \right) + C$$

$$y = \frac{x^2}{2} + \frac{x}{2} - \frac{1}{4} + Ce^{-2x}$$

$$\boxed{y = \frac{x^2}{2} + \frac{x}{2} - \frac{1}{4} + Ce^{-2x}}$$

Operación auxiliar

$$\int e^{2x}(x^2 + 2x) dx = \int x^2 e^{2x} dx + 2 \int x e^{2x} dx$$

$$\int x^2 e^{2x} dx \quad u = x^2, \quad dv = e^{2x} dx$$

$$= \frac{x^2}{2} e^{2x} - \int x e^{2x} dx$$

$$\int x e^{2x} dx \quad u = x, \quad dv = e^{2x} dx$$

$$= \frac{x}{2} e^{2x} - \frac{1}{4} e^{2x}$$

$$\int x^2 e^{2x} dx = \frac{x^2}{2} e^{2x} - \frac{x}{2} e^{2x} + \frac{1}{4} e^{2x}$$

$$2 \int x e^{2x} dx = x e^{2x} - \frac{1}{2} e^{2x}$$

$$\int e^{2x}(x^2 + 2x) dx = e^{2x} \left(\frac{x^2}{2} + \frac{x}{2} - \frac{1}{4} \right)$$

Ejercicio 3

$$\frac{dy}{dx} = \frac{y(x^2 - 1)}{y + 1}$$

$$\frac{y + 1}{y} dy = (x^2 - 1) dx$$

$$\left(1 + \frac{1}{y}\right) dy = (x^2 - 1) dx$$

$$\int \left(1 + \frac{1}{y}\right) dy = \int (x^2 - 1) dx$$

$$y + \ln |y| = \frac{x^3}{3} - x + C$$

$$\boxed{y + \ln |y| = \frac{x^3}{3} - x + C}$$

Ejercicio 4

$$y' + y = e^{-x}$$

$$\mu = e^{\int 1 dx}$$

$$\mu = e^x$$

$$e^x y' + e^x y = 1$$

$$\frac{d}{dx} (ye^x) = 1$$

$$ye^x = \int 1 dx + C$$

$$ye^x = x + C$$

$$y = (x + C)e^{-x}$$

$$\boxed{y = (x + C)e^{-x}}$$

Ejercicio 5

$$y' - y \cot x = 2x - x^2 \cot x$$

$$\mu = e^{\int -\cot x \, dx}$$

$$\mu = e^{-\ln |\operatorname{sen} x|} = \csc x$$

$$\csc x \, y' - \csc x \cot x \, y = 2x \csc x - x^2 \csc x \cot x$$

$$\frac{d}{dx} (y \csc x) = 2x \csc x - x^2 \csc x \cot x$$

$$y \csc x = \int (2x \csc x - x^2 \csc x \cot x) \, dx + C$$

$$y \csc x = x^2 \csc x + C$$

$$y = x^2 + C \operatorname{sen} x$$

$$\boxed{y = x^2 + C \operatorname{sen} x}$$

Operación auxiliar

$$\frac{d}{dx} (x^2 \csc x) = 2x \csc x - x^2 \csc x \cot x$$

$$\int (2x \csc x - x^2 \csc x \cot x) \, dx = x^2 \csc x$$

Ejercicio 6

$$8xy' - y = -\frac{1}{y^3 \sqrt{x+1}}$$

$$y' - \frac{1}{8x}y = -\frac{1}{8x \sqrt{x+1}}y^{-3}$$

$$n = -3, \quad v = y^4, \quad v' = 4y^3 y'$$

$$4y^3 y' - \frac{1}{2x}y^4 = -\frac{1}{2x \sqrt{x+1}}$$

$$v' - \frac{1}{2x}v = -\frac{1}{2x \sqrt{x+1}}$$

$$\mu = e^{\int -\frac{1}{2x} \, dx} = x^{-1/2}$$

$$x^{-1/2}v' - \frac{1}{2}x^{-3/2}v = -\frac{1}{2}x^{-3/2}\frac{1}{\sqrt{x+1}}$$

$$\frac{d}{dx}(x^{-1/2}v) = -\frac{1}{2}\frac{1}{x^{3/2}\sqrt{x+1}}$$

$$x^{-1/2}v = -\frac{1}{2}\int \frac{dx}{x^{3/2}\sqrt{x+1}} + C$$

$$x^{-1/2}v = \frac{\sqrt{x+1}}{\sqrt{x}} + C$$

$$v = \sqrt{x+1} + C\sqrt{x}$$

$$y^4 = \sqrt{x+1} + C\sqrt{x}$$

$$\boxed{y^4 = \sqrt{x+1} + C\sqrt{x}}$$

Operación auxiliar

$$I = \int \frac{dx}{x^{3/2}\sqrt{x+1}}$$

$$t = \sqrt{x}, \quad x = t^2, \quad dx = 2t \, dt$$

$$I = \int \frac{2t \, dt}{t^3\sqrt{t^2+1}} = 2 \int \frac{dt}{t^2\sqrt{t^2+1}}$$

$$t = \tan \theta, \quad dt = \sec^2 \theta \, d\theta$$

$$I = 2 \int \frac{\sec^2 \theta \, d\theta}{\tan^2 \theta \sec \theta} = 2 \int \frac{\cos \theta}{\sin^2 \theta} \, d\theta$$

$$I = -2 \csc \theta$$

$$\sin \theta = \frac{\sqrt{x}}{\sqrt{x+1}}, \quad \csc \theta = \frac{\sqrt{x+1}}{\sqrt{x}}$$

$$I = -2 \frac{\sqrt{x+1}}{\sqrt{x}}$$

$$-\frac{1}{2}I = \frac{\sqrt{x+1}}{\sqrt{x}}$$

II) ECUACIONES DE BERNOULLI

Ejercicio 7

$$y' - \frac{2}{x}y = \frac{\ln x}{x}y^2$$

$$n = 2, \quad v = y^{-1}, \quad v' = -y^{-2}y'$$

$$-y^{-2}y' + \frac{2}{x}y^{-1} = -\frac{\ln x}{x}$$

$$v' + \frac{2}{x}v = -\frac{\ln x}{x}$$

$$\mu = e^{\int \frac{2}{x} dx} = x^2$$

$$x^2v' + 2xv = -x \ln x$$

$$\frac{d}{dx}(x^2v) = -x \ln x$$

$$x^2v = -\int x \ln x dx + C$$

$$x^2v = -\frac{x^2}{2} \ln x + \frac{x^2}{4} + C$$

$$v = -\frac{1}{2} \ln x + \frac{1}{4} + \frac{C}{x^2}$$

$$\frac{1}{y} = -\frac{1}{2} \ln x + \frac{1}{4} + \frac{C}{x^2}$$

$$\boxed{\frac{1}{y} = -\frac{1}{2} \ln x + \frac{1}{4} + \frac{C}{x^2}}$$

Operación auxiliar

$$\int x \ln x dx$$

$$u = \ln x, \quad du = \frac{dx}{x}, \quad dv = x dx, \quad v = \frac{x^2}{2}$$

$$= \frac{x^2}{2} \ln x - \int \frac{x}{2} dx$$

$$= \frac{x^2}{2} \ln x - \frac{x^2}{4}$$

Ejercicio 8

$$y' + xy = x^3 y^3$$

$$n = 3, \quad v = y^{-2}, \quad v' = -2y^{-3}y'$$

$$-2y^{-3}y' - 2xy^{-2} = -2x^3$$

$$v' - 2xv = -2x^3$$

$$\mu = e^{\int -2x \, dx} = e^{-x^2}$$

$$e^{-x^2}v' - 2xe^{-x^2}v = -2x^3e^{-x^2}$$

$$\frac{d}{dx} \left(e^{-x^2}v \right) = -2x^3e^{-x^2}$$

$$e^{-x^2}v = \int -2x^3e^{-x^2} \, dx + C$$

$$e^{-x^2}v = e^{-x^2}(x^2 + 1) + C$$

$$v = x^2 + 1 + Ce^{x^2}$$

$$y^{-2} = x^2 + 1 + Ce^{x^2}$$

$$\boxed{\frac{1}{y^2} = x^2 + 1 + Ce^{x^2}}$$

Operación auxiliar

$$\int -2x^3e^{-x^2} \, dx$$

$$u = x^2, \quad du = 2x \, dx, \quad x^3 \, dx = u \frac{du}{2}$$

$$= \int -ue^{-u} \, du$$

$$= ue^{-u} + e^{-u} = e^{-u}(u + 1)$$

$$= e^{-x^2}(x^2 + 1)$$

Ejercicio 9

$$y' + y = e^x y^2$$

$$n = 2, \quad v = y^{-1}, \quad v' = -y^{-2} y'$$

$$-y^{-2} y' - y^{-1} = -e^x$$

$$v' - v = -e^x$$

$$\mu = e^{\int -1 dx} = e^{-x}$$

$$e^{-x} v' - e^{-x} v = -1$$

$$\frac{d}{dx} (e^{-x} v) = -1$$

$$e^{-x} v = -x + C$$

$$v = -x e^x + C e^x = e^x (C - x)$$

$$\frac{1}{y} = e^x (C - x)$$

$$\boxed{\frac{1}{y} = e^x (C - x)}$$

Ejercicio 10

$$\frac{1}{y'} = xy + x^2 y^3$$

$$\frac{dx}{dy} = xy + x^2 y^3$$

$$\frac{dx}{dy} - yx = y^3 x^2$$

$$n = 2, \quad v = x^{-1}, \quad v' = -x^{-2} x'$$

$$-x^{-2} \frac{dx}{dy} + yx^{-1} = -y^3$$

$$v' + yv = -y^3$$

$$\mu = e^{\int y \, dy} = e^{y^2/2}$$

$$e^{y^2/2}v' + ye^{y^2/2}v = -y^3e^{y^2/2}$$

$$\frac{d}{dy} \left(e^{y^2/2}v \right) = -y^3e^{y^2/2}$$

$$e^{y^2/2}v = \int -y^3e^{y^2/2} \, dy + C$$

$$e^{y^2/2}v = e^{y^2/2}(2 - y^2) + C$$

$$v = 2 - y^2 + Ce^{-y^2/2}$$

$$\frac{1}{x} = 2 - y^2 + Ce^{-y^2/2}$$

$$\boxed{\frac{1}{x} = 2 - y^2 + Ce^{-y^2/2}}$$

Operación auxiliar

$$\int -y^3e^{y^2/2} \, dy$$

$$u = \frac{y^2}{2}, \, du = y \, dy, \, y^3 \, dy = 2u \, du$$

$$= \int -2ue^u \, du = -2e^u(u - 1)$$

$$= 2e^u(1 - u) = e^{y^2/2}(2 - y^2)$$

Ejercicio 11

$$xy' + \frac{y}{\ln x} = \frac{x^2}{y^2}$$

$$y' + \frac{1}{x \ln x}y = xy^{-2}$$

$$n = -2, \quad v = y^3, \quad v' = 3y^2y'$$

$$3y^2y' + \frac{3}{x \ln x}y^3 = 3x$$

$$v' + \frac{3}{x \ln x}v = 3x$$

$$\mu = e^{\int \frac{3}{x \ln x} dx} = (\ln x)^3$$

$$(\ln x)^3 v' + \frac{3(\ln x)^2}{x} v = 3x(\ln x)^3$$

$$\frac{d}{dx} [(\ln x)^3 v] = 3x(\ln x)^3$$

$$(\ln x)^3 v = \int 3x(\ln x)^3 dx + C$$

$$(\ln x)^3 v = \frac{3x^2}{2}(\ln x)^3 - \frac{9x^2}{4}(\ln x)^2 + \frac{9x^2}{4} \ln x - \frac{9x^2}{8} + C$$

$$v = \frac{\frac{3x^2}{2}(\ln x)^3 - \frac{9x^2}{4}(\ln x)^2 + \frac{9x^2}{4} \ln x - \frac{9x^2}{8} + C}{(\ln x)^3}$$

$$y^3 = \frac{\frac{3x^2}{2}(\ln x)^3 - \frac{9x^2}{4}(\ln x)^2 + \frac{9x^2}{4} \ln x - \frac{9x^2}{8} + C}{(\ln x)^3}$$

$$y^3 = \frac{\frac{3x^2}{2}(\ln x)^3 - \frac{9x^2}{4}(\ln x)^2 + \frac{9x^2}{4} \ln x - \frac{9x^2}{8} + C}{(\ln x)^3}$$

Operación auxiliar

$$\int 3x(\ln x)^3 dx = 3 \int x(\ln x)^3 dx$$

$$\int x(\ln x)^3 dx$$

$$u = (\ln x)^3, \quad dv = x dx \quad \Rightarrow \quad du = 3(\ln x)^2 \frac{1}{x} dx, \quad v = \frac{x^2}{2}$$

$$= \frac{x^2}{2}(\ln x)^3 - \frac{3}{2} \int x(\ln x)^2 dx$$

$$\int x(\ln x)^2 dx \quad u = (\ln x)^2, \quad dv = x dx$$

$$= \frac{x^2}{2}(\ln x)^2 - \int x \ln x dx$$

$$\int x \ln x dx = \frac{x^2}{2} \ln x - \frac{x^2}{4}$$

$$\int x(\ln x)^2 dx = \frac{x^2}{2}(\ln x)^2 - \frac{x^2}{2} \ln x + \frac{x^2}{4}$$

$$\int x(\ln x)^3 dx = \frac{x^2}{2}(\ln x)^3 - \frac{3x^2}{4}(\ln x)^2 + \frac{3x^2}{4} \ln x - \frac{3x^2}{8}$$

$$\int 3x(\ln x)^3 dx = \frac{3x^2}{2}(\ln x)^3 - \frac{9x^2}{4}(\ln x)^2 + \frac{9x^2}{4} \ln x - \frac{9x^2}{8}$$

Ejercicio 12

$$y' = \frac{2xy}{x^2 - y^2 - a^2}$$

$$\frac{dx}{dy} = \frac{x^2 - y^2 - a^2}{2xy} = \frac{x}{2y} - \frac{y^2 + a^2}{2y}x^{-1}$$

$$\frac{dx}{dy} - \frac{1}{2y}x = -\frac{y^2 + a^2}{2y}x^{-1}$$

$$n = -1, \quad v = x^2, \quad v' = 2x \frac{dx}{dy}$$

$$2x \frac{dx}{dy} - \frac{1}{y}x^2 = -\frac{y^2 + a^2}{y}$$

$$v' - \frac{1}{y}v = -\frac{y^2 + a^2}{y}$$

$$\mu = e^{\int -\frac{1}{y} dy} = \frac{1}{y}$$

$$\frac{1}{y}v' - \frac{1}{y^2}v = -1 - \frac{a^2}{y^2}$$

$$\frac{d}{dy} \left(\frac{v}{y} \right) = -1 - \frac{a^2}{y^2}$$

$$\frac{v}{y} = \int \left(-1 - \frac{a^2}{y^2} \right) dy = -y + \frac{a^2}{y} + C$$

$$v = -y^2 + a^2 + Cy$$

$$x^2 = -y^2 + Cy + a^2$$

$$\boxed{x^2 + y^2 - Cy = a^2}$$

Ejercicio 13

$$\frac{dy}{dx} = \frac{4 \operatorname{sen}^2 y}{x^5 + x \tan y}$$

$$\frac{dx}{dy} = \frac{x^5 + x \tan y}{4 \operatorname{sen}^2 y} = \frac{1}{4 \operatorname{sen}^2 y}x^5 + \frac{\tan y}{4 \operatorname{sen}^2 y}x$$

$$\frac{dx}{dy} - \frac{\tan y}{4 \operatorname{sen}^2 y}x = \frac{1}{4 \operatorname{sen}^2 y}x^5$$

$$n = 5, \quad v = x^{-4}, \quad v' = -4x^{-5} \frac{dx}{dy}$$

$$-4x^{-5} \frac{dx}{dy} + \frac{\tan y}{\sin^2 y} x^{-4} = -\frac{1}{\sin^2 y}$$

$$\frac{\tan y}{\sin^2 y} = \frac{1}{\sin y \cos y}$$

$$v' + \frac{1}{\sin y \cos y} v = -\csc^2 y$$

$$\mu = e^{\int \frac{dy}{\sin y \cos y}} = e^{\ln |\tan y|} = \tan y$$

$$\tan y v' + \frac{1}{\cos^2 y} v = -\csc^2 y \tan y$$

$$\frac{d}{dy} (v \tan y) = -\frac{1}{\sin y \cos y}$$

$$v \tan y = \int -\frac{dy}{\sin y \cos y} + C = -\ln |\tan y| + C$$

$$v = \frac{C - \ln |\tan y|}{\tan y}$$

$$x^{-4} = \frac{C - \ln |\tan y|}{\tan y}$$

$$\boxed{x^4 = \frac{\tan y}{C - \ln |\tan y|}}$$

III) ECUACIONES DE RICCATI

Ejercicio 14

$$y' = xy^2 - (2x - 1)y + x - 1, \quad y_1 = 1$$

$$y = 1 + \frac{1}{v}, \quad y' = -\frac{v'}{v^2}$$

$$-\frac{v'}{v^2} = x \left(1 + \frac{2}{v} + \frac{1}{v^2}\right) - (2x - 1) \left(1 + \frac{1}{v}\right) + x - 1$$

$$-\frac{v'}{v^2} = \frac{1}{v} + \frac{x}{v^2}$$

$$v' = -v - x$$

$$v' + v = -x$$

$$\mu = e^{\int 1 dx} = e^x$$

$$e^x v' + e^x v = -x e^x$$

$$\frac{d}{dx}(e^x v) = -x e^x$$

$$e^x v = -\int x e^x dx + C$$

$$e^x v = -e^x(x-1) + C$$

$$v = 1 - x + C e^{-x}$$

$$y = 1 + \frac{1}{1 - x + C e^{-x}}$$

$$\boxed{y = 1 + \frac{1}{1 - x + C e^{-x}}}$$

Operación auxiliar

$$\int x e^x dx$$

$$u = x, dv = e^x dx \Rightarrow du = dx, v = e^x$$

$$= x e^x - \int e^x dx = x e^x - e^x = e^x(x-1)$$

Ejercicio 15

$$y' = -8xy^2 + (16x^2 + 4x)y - 8x^3 - 4x^2 + 1, \quad y_1 = x$$

$$y = x + \frac{1}{v}, \quad y' = 1 - \frac{v'}{v^2}$$

$$1 - \frac{v'}{v^2} = -8x \left(x^2 + \frac{2x}{v} + \frac{1}{v^2} \right) + (16x^2 + 4x) \left(x + \frac{1}{v} \right) - 8x^3 - 4x^2 + 1$$

$$1 - \frac{v'}{v^2} = 1 + \frac{4x}{v} - \frac{8x}{v^2}$$

$$-\frac{v'}{v^2} = \frac{4x}{v} - \frac{8x}{v^2}$$

$$v' + 4xv = 8x$$

$$\mu = e^{\int 4x dx} = e^{2x^2}$$

$$e^{2x^2} v' + 4xe^{2x^2} v = 8xe^{2x^2}$$

$$\frac{d}{dx} \left(e^{2x^2} v \right) = 8xe^{2x^2}$$

$$e^{2x^2} v = \int 8xe^{2x^2} dx + C$$

$$e^{2x^2} v = 2e^{2x^2} + C$$

$$v = 2 + Ce^{-2x^2}$$

$$y = x + \frac{1}{2 + Ce^{-2x^2}}$$

$$\boxed{y = x + \frac{1}{2 + Ce^{-2x^2}}}$$

Operación auxiliar

$$\int 8xe^{2x^2} dx$$

$$u = 2x^2, \quad du = 4x \, dx, \quad 8x \, dx = 2 \, du$$

$$= 2 \int e^u \, du = 2e^u = 2e^{2x^2}$$